Omid Mokhtari

Ph.D. Student in Electrical Engineering, Power Systems

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Research Interests

Optimal network reduction for transmission and distribution networks; scalable optimization (decomposition, parallel and GPU computing); voltage estimation and sensor placement with sparse measurements; power system planning and operation.

Education

Ph.D. **University of Vermont**, Electrical Engineering — Burlington, VT, USA — Aug 2023 – present
- Expected completion: Fall 2027
- Advisors: Dr. Samuel Chevalier and Dr. Mads R. Almassalkhi
- Research: scalable optimization and network reduction methods for electric power networks

M.Sc. **Sharif University of Technology**, Electrical Engineering, Power Systems — Tehran, Iran — Sept 2017 – Jan 2020
- Thesis: Co-planning of Multiple Merchant Energy Storage Systems and Transmission Systems
- Advisor: Prof. Seyed Hamid Hosseini

B.Sc. **Imam Khomeini International University**, Electrical Engineering, Power Systems — Qazvin, Iran — Sept 2012 – July 2017
- Final project: Investigation of Wind Turbines Dynamic Behavior During a Fault, Using DIgSILENT
- Advisor: Dr. Navid Ghaffarzadeh

Journal Papers

Optimal Kron-based Reduction of Networks (Opti-KRON) for Three-phase Distribution Feeders — 2027
Omid Mokhtari, Samuel Chevalier, Mads R. Almassalkhi
10.1016/j.epsr.2026.113615 (Electric Power Systems Research, vol. 263, 113615. Presented at PSCC 2026, Limassol, Cyprus)

Structure-preserving Optimal Kron-based Reduction of Radial Distribution Networks — 2026
Omid Mokhtari, Samuel Chevalier, Mads R. Almassalkhi
10.1109/TPWRS.2026.3726028 (IEEE Transactions on Power Systems (early access))

Preprints

Data-Driven Power Flow for Radial Distribution Networks with Sparse Real-Time Data — Apr 2026
Oleksii Molodchyk, **Omid Mokhtari**, Samuel Chevalier, Mads R. Almassalkhi, Timm Faulwasser
arxiv.org/abs/2604.08403 (arXiv preprint arXiv:2604.08403)

Enhancing Scalability of Optimal Kron-based Reduction of Networks (Opti-KRON) via Decomposition with Community Detection — July 2024
Omid Mokhtari, Samuel Chevalier, Mads R. Almassalkhi
arxiv.org/abs/2407.02679 (arXiv preprint arXiv:2407.02679)

Presentations

Optimal Kron-based Reduction of Networks (Opti-KRON) for Three-phase Distribution Feeders — Limassol, Cyprus, June 2026
Power Systems Computation Conference (PSCC 2026)

Structure-preserving Optimal Kron-based Reduction of Radial Distribution Networks — Boulder, CO, Sept 2026
PowerUp 2026 (poster)

Optimal Kron-based Reduction of Networks (Opti-KRON) for Three-phase Distribution Feeders — Burlington, VT, 2026
Student Research Conference, University of Vermont (poster)

Network Reduction and Radialization — Burlington, VT, 2025
Student Research Conference, University of Vermont (poster)

Optimal Network Reduction — Seattle, WA, July 2024
IEEE Power & Energy Society General Meeting (poster)

Community Detection Based Network Reduction — Burlington, VT, 2024
Student Research Conference, University of Vermont (poster)

Research Experience

University of Vermont, Department of Electrical and Biomedical Engineering, Graduate Research Assistant — Burlington, VT, USA, Aug 2023 – present

- Developed Opti-KRON, an optimization-based network reduction method for radial and unbalanced three-phase distribution feeders; removed over 94% of nodes on a 3,499-bus utility feeder with voltage error below 0.002 p.u.
- Accelerated three-phase network reduction on GPUs (up to 15x faster than CPU) and scaled reduction to large transmission networks through community-detection decomposition.
- Co-developed a data-driven power flow and sensor placement method that estimates voltages with errors below 0.001 p.u. when only about a quarter of locations have sensors.

Sharif University of Technology, Research Assistant — Tehran, Iran, Sept 2017 – Jan 2020

- Developed planning methods for co-planning merchant energy storage systems with transmission systems.

Professional Service

Reviewer: IEEE Transactions on Power Systems; Electric Power Systems Research; American Control Conference (ACC)

Technical Skills

Programming: Julia, Python, MATLAB, GAMS

Software: DIgSILENT PowerFactory, LaTeX